GroPanda System Design



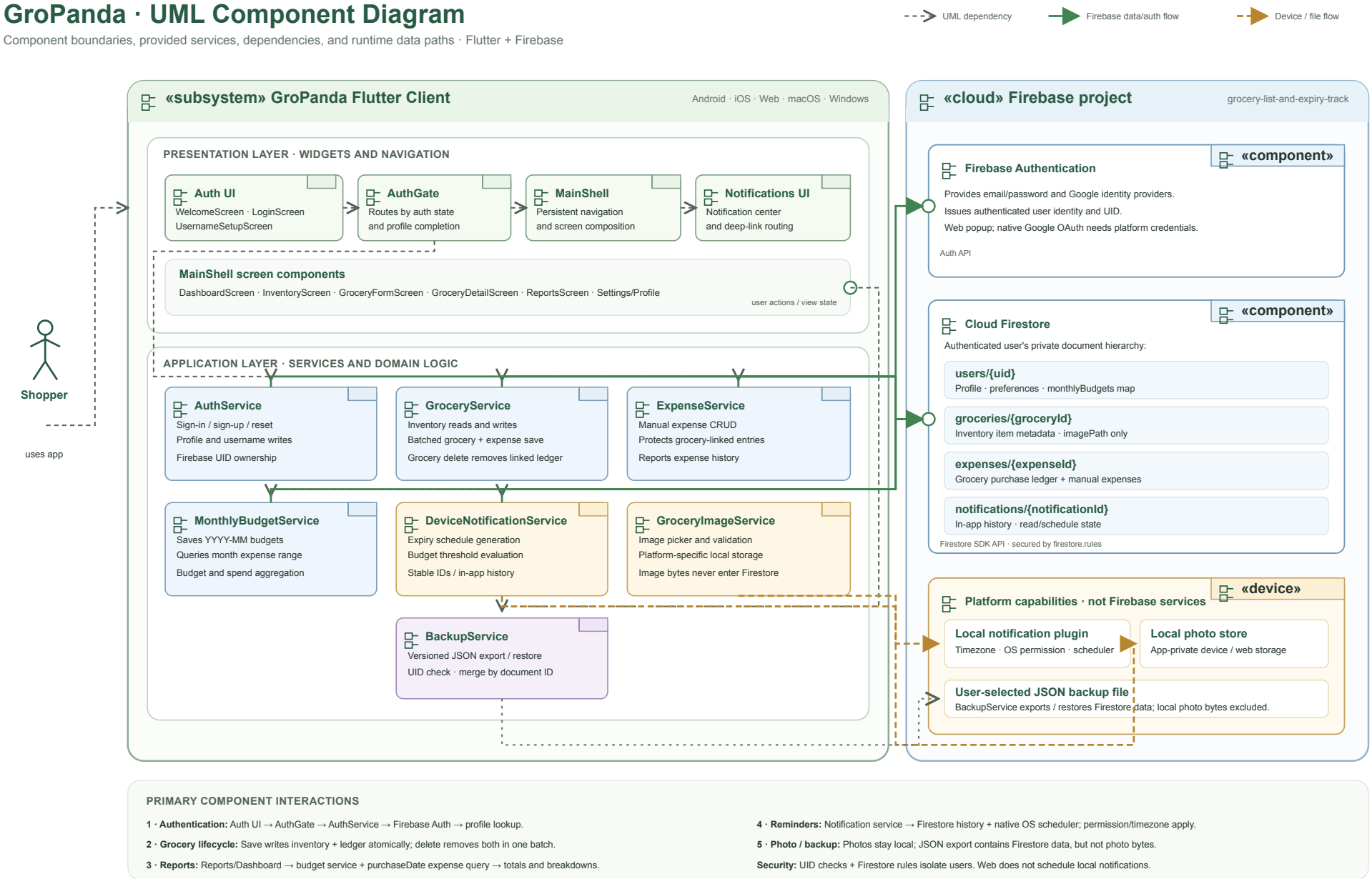
This document describes the current Flutter and Firebase implementation. The database is Cloud Firestore, a document database; the schema below therefore describes collections and documents rather than relational tables.

Printable copy: SYSTEM_DESIGN.pdf. Regenerate it after editing this document by running `python3 docs/export_system_design_pdf.py` from the repository root. The exporter uses local Google Chrome and Python's standard library; it does not upload the document.

1. System architecture

GroPanda · UML Component Diagram

Component boundaries, provided services, dependencies, and runtime data paths · Flutter + Firebase



PRIMARY COMPONENT INTERACTIONS

1 · Authentication: Auth UI → AuthGate → AuthService → Firebase Auth → profile lookup.

2 · Grocery lifecycle: Save writes inventory + ledger atomically; delete removes both in one batch.

3 · Reports: Reports/Dashboard → budget service + purchaseDate expense query → totals and breakdowns.

4 · Reminders: Notification service → Firestore history + native OS scheduler; permission/timezone apply.

5 · Photo / backup: Photos stay local; JSON export contains Firestore data, but not photo bytes.

Security: UID checks + Firestore rules isolate users. Web does not schedule local notifications.

Detailed system architecture showing app screens, services, Firebase collections, device-local features, and end-to-end flows

This uses a UML/Lucidchart-style component notation: component glyphs and component boundaries show deployable/runtime responsibilities; dashed dependency arrows show which component calls another; labeled Firebase and device flows distinguish cloud data access from local capabilities. The diagram's interaction notes summarize the sign-in, grocery, reporting, reminder, photo, and backup paths. The Flutter client talks directly to Firebase through the Firebase SDK; there is no separate application server.

Main data flows

1. Firebase Authentication signs the user in. The auth gate loads

`users/{uid}` and sends users without a username to username setup.

1. The main shell presents Dashboard, Inventory, Add, Reports, and Settings.

Screen-level services access records beneath the current user's UID.

1. Saving a grocery writes the grocery document and its expense-ledger entry

together in a Firestore batch. Updating a grocery updates both documents. Deleting a grocery atomically deletes both the inventory item and its linked purchase expense; unrelated manual expenses are retained.

1. The app schedules expiry alerts on supported native platforms using device

local notifications. Notification history and budget alerts are stored in Firestore. Web does not schedule device notifications.

1. Grocery image bytes remain in app-private local storage. Firestore stores

only the local `imagePath`; images do not sync to other devices and are not included in JSON backups.

1. Backup export and restore are initiated by the user. Restore merges matching

document IDs and only accepts a backup belonging to the signed-in UID.

2. Firestore data model

All user-owned data is rooted at the authenticated UID:

```
users/{uid}
  groceries/{groceryId}
  expenses/{expenseId}
  notifications/{notificationId}
```

`{uid}` is the Firebase Authentication UID. The security rules in `firestore.rules` allow a signed-in user to read and write only their own user document and subcollections.

`users/{uid}` — profile and preferences

Field	Type	Description
`uid`	string	Firebase Authentication UID.
`email`	string, optional	Account email.

Field	Type	Description
`displayName`	string, optional	Provider display name.
`photoURL`	string, optional	Provider profile image URL.
`username`	string, optional until onboarding	Public-facing username; 3–20 letters, numbers, or underscores.
`usernameLower`	string, optional	Lowercase username value saved for normalized lookup.
`profileCompleted`	boolean	Whether username setup has completed.
`createdAt`	timestamp	Account profile creation time.
`lastLoginAt`	timestamp, optional	Last Google sign-in time when available.
`updatedAt`	timestamp, optional	Last profile or budget update.
`monthlyBudget`	number, optional	Legacy/current-month budget value.
`monthlyBudgets`	map of `YYYY-MM` to number, optional	Per-month budget amounts.
`notificationEnabled`	boolean	Master notification preference; defaults to true.
`notificationTime`	string (`HH:mm`)	Local preferred notification time; defaults to `08:00` .
`expiryReminder7Days`	boolean	Enable reminder seven days before expiry.
`expiryReminder3Days`	boolean	Enable reminder three days before expiry.
`expiryReminder1Day`	boolean	Enable reminder one day before expiry.
`expiryReminderOnExpiry`	boolean	Enable reminder on the expiry date.
`budgetAlert70`	boolean	Enable 70% budget threshold alert.
`budgetAlert90`	boolean	Enable 90% budget threshold alert.
`budgetAlert100`	boolean	Enable 100% budget threshold alert.
`budgetAlertOver100`	boolean	Enable over-budget alert.

Budget map example:

```
{
  "monthlyBudget": 12000,
  "monthlyBudgets": {
    "2026-10": 12000,
    "2026-11": 13500
  }
}
```

`users/{uid}/groceries/{groceryId}` — inventory item

Field	Type	Description
`name`	string	Grocery name.
`category`	string	Category such as Fruits, Dairy, Pantry, or Other.
`quantity`	number	Amount owned.

Field	Type	Description
`unitPrice`	number	Price per unit; defaults to zero if not supplied.
`unit`	string	Unit such as pieces, packs, kg, or liters.
`purchaseDate`	timestamp	Date purchased.
`expirationDate`	timestamp	Expiry date; used for ordering and reminders.
`storageLocation`	string	Location such as fridge, freezer, or pantry.
`notes`	string	Optional free-form notes (stored as an empty string when blank).
`createdAt`	timestamp	Creation time, set by the server on add.
`updatedAt`	timestamp	Last update time, set by the server.
`imagePath`	string or null	Device-local photo path; photo bytes are not in Firestore.

The document ID is generated by Firestore before the first write and is reused for updates. The item total cost is calculated as `quantity * unitPrice`; it is not stored as a separate grocery field.

`users/{uid}/expenses/{expenseId}` — spending ledger

Grocery-linked expenses use the grocery document ID as `expenseId`. They are written atomically with the grocery record, so editing the item updates its ledger entry. Removing the inventory item deletes its linked expense in the same batch; unrelated manual expenses are retained.

Field	Type	Description
`userId`	string	Owning Firebase UID.
`groceryId`	string, optional	Present for grocery-linked purchases; absent for manual expenses.
`name`	string	Expense name; maintained for compatibility/display.
`description`	string	User-facing expense description.
`amount`	number	Total expense amount.
`category`	string	Expense category.
`date`	timestamp	Expense date; stored for both expense kinds.
`purchaseDate`	timestamp	Date used by monthly report and budget queries.
`source`	string, optional	Manual expenses use `manual`; grocery-linked records are identified by `groceryId`.
`createdAt`	timestamp	Creation time.
`updatedAt`	timestamp	Last update time.

Manual expenses receive an automatically generated document ID and can be edited or deleted from Reports. Grocery-linked expenses must be changed through their grocery item.

`users/{uid}/notifications/{notificationId}` — in-app notification history

Notification IDs are deterministic for expiry reminders and budget thresholds, which prevents duplicate records for the same event.

Field	Type	Description
`userId`	string	Owning Firebase UID.
`type`	string	`expiry` or `budget`.
`title`	string, optional	Stored on budget notifications; the UI supplies a fallback title otherwise.
`message`	string	Notification content shown in the app.
`createdAt`	timestamp	Notification record creation time.
`read`	boolean	Whether the user has read the notification.
`itemId`	string, expiry only	Grocery document ID to open.
`itemName`	string, expiry only	Grocery name at scheduling time.
`expiryDate`	timestamp, expiry only	Expiration date the reminder refers to.
`scheduledAt`	timestamp, expiry only	Local reminder target time represented as a Firestore timestamp.
`reminderScheduled`	boolean, expiry only	Whether a local reminder was scheduled.
`reminderSent`	boolean, expiry only	Whether the reminder has already been delivered/handled.
`month`	string (`YYYY-MM`), budget only	Budget month.
`threshold`	number, budget only	Threshold percentage (70, 90, 100, or 101 for exceeded).
`budgetAmount`	number, budget only	Budget configured for the month.
`amountSpent`	number, budget only	Spend at threshold evaluation time.
`percentage`	number, budget only	Percentage of budget spent.
`phoneNotificationShown`	boolean, budget only	Whether a native alert was shown.

3. Queries and operational notes

- Inventory streams are ordered by `expirationDate`.
- The notification center orders by `createdAt` descending and reads at most

100 records.

- Monthly reports and budget alerts filter expenses by `purchaseDate` between

the first day of the month and the first day of the next month.

- Firestore automatic single-field indexing is used by these current queries.

Revisit index requirements if query shapes change.

- All timestamps are Firestore `Timestamp` values. Budget map keys use

zero-padded `YYYY-MM`.

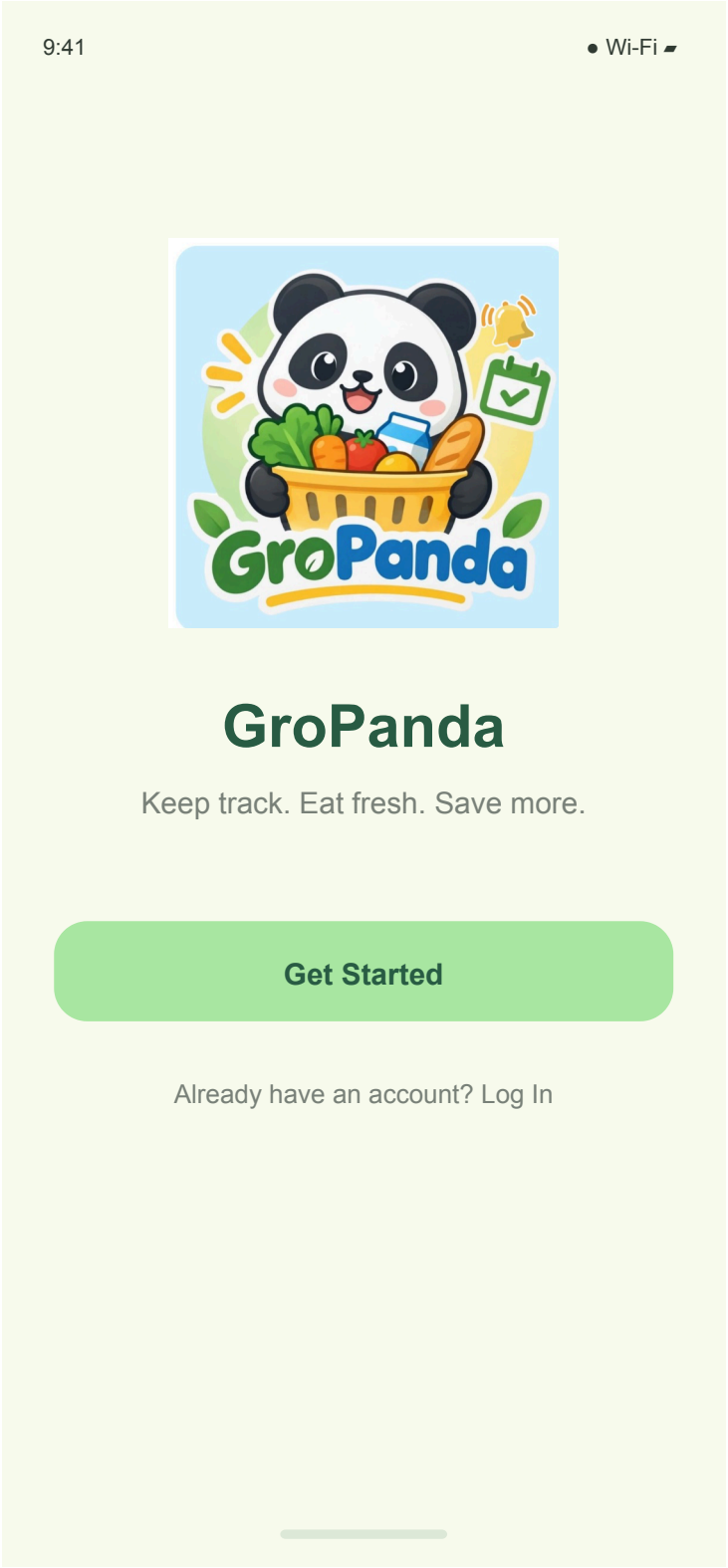
- The current rules enforce owner UID isolation, but they do not validate field

types, required fields, or value ranges. Add Firestore field validation before enabling untrusted clients or introducing additional write paths.

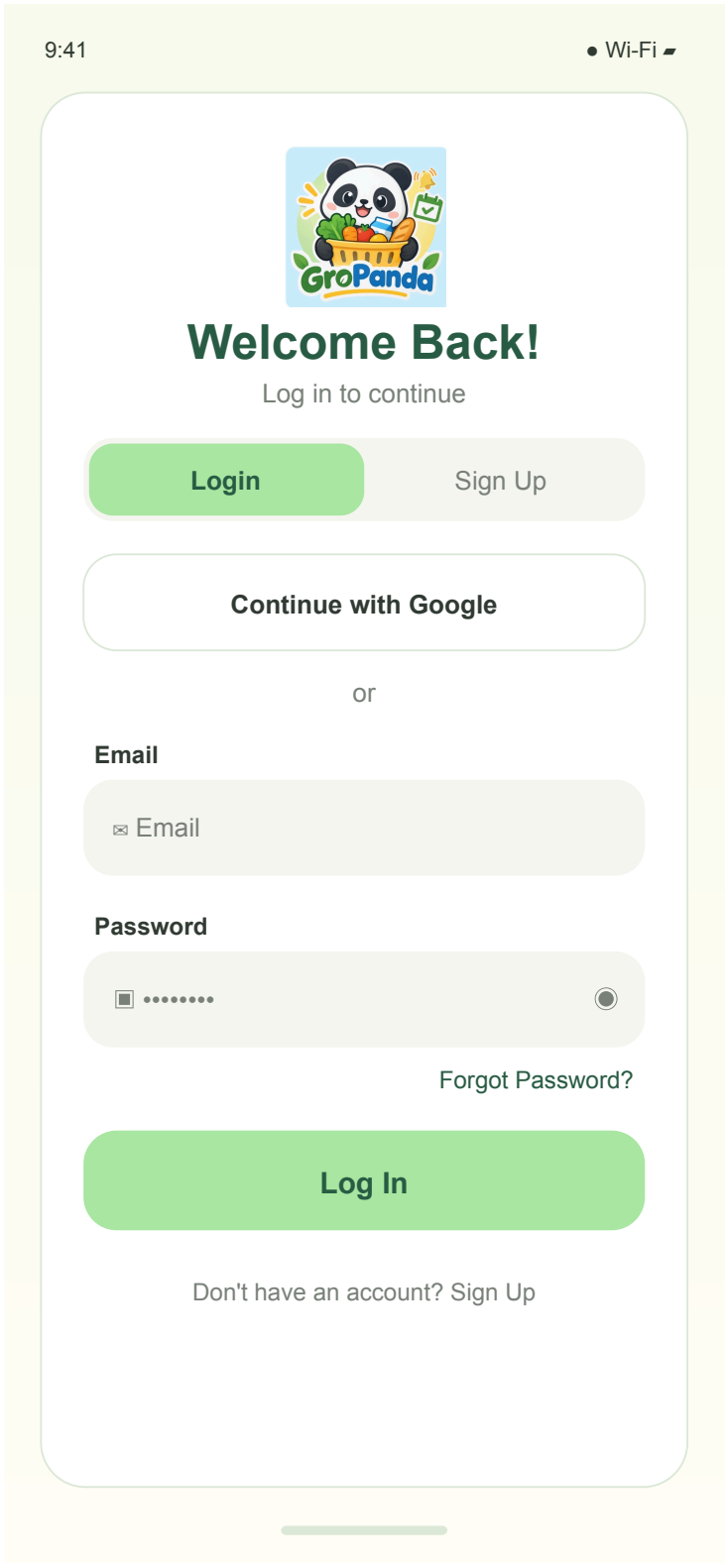
4. Visual wireframes

These mobile wireframes have been revised to reflect the current screen structure, controls, content, and theme in the Flutter implementation. They remain illustrative rather than pixel-perfect renderings: data is representative and long pages show only their initial viewport.

Welcome and authentication

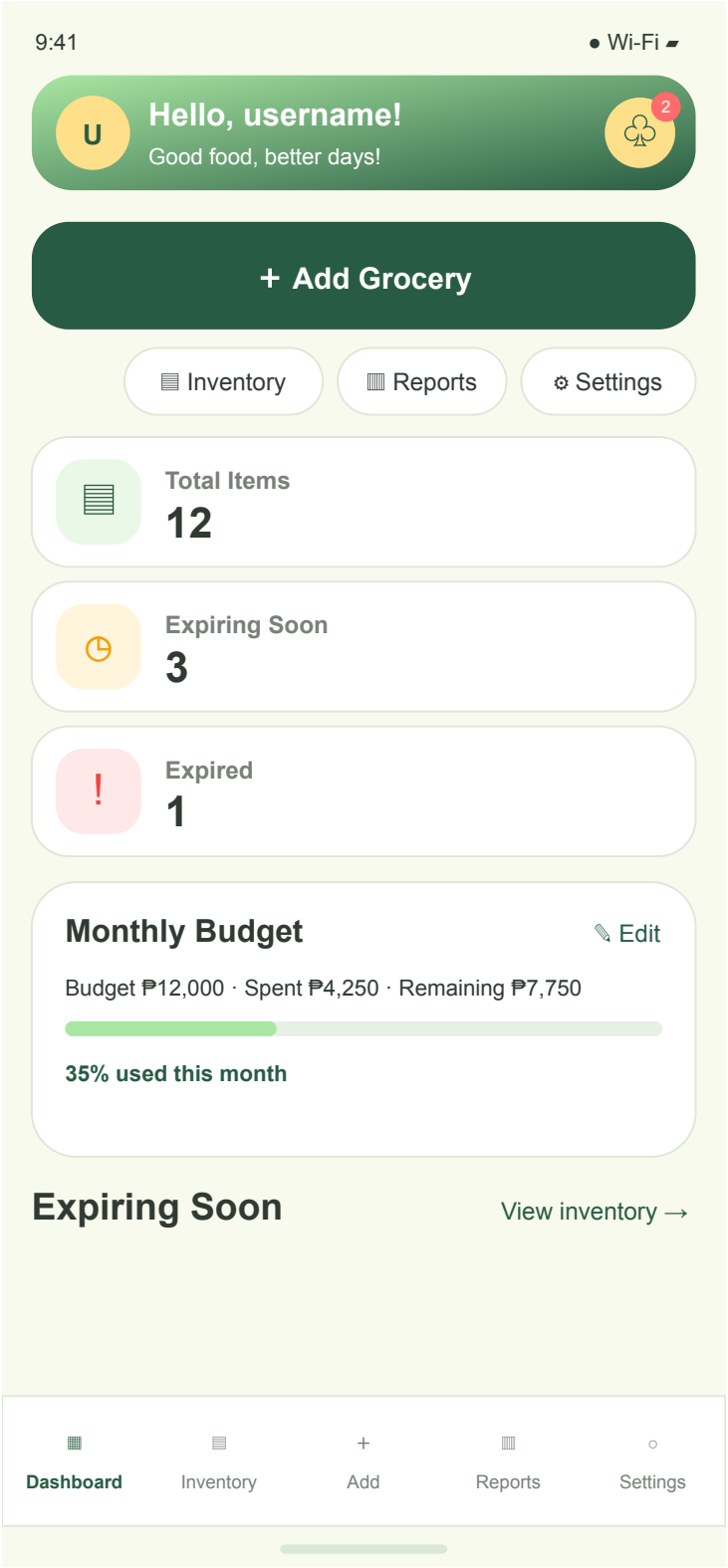


Welcome screen wireframe

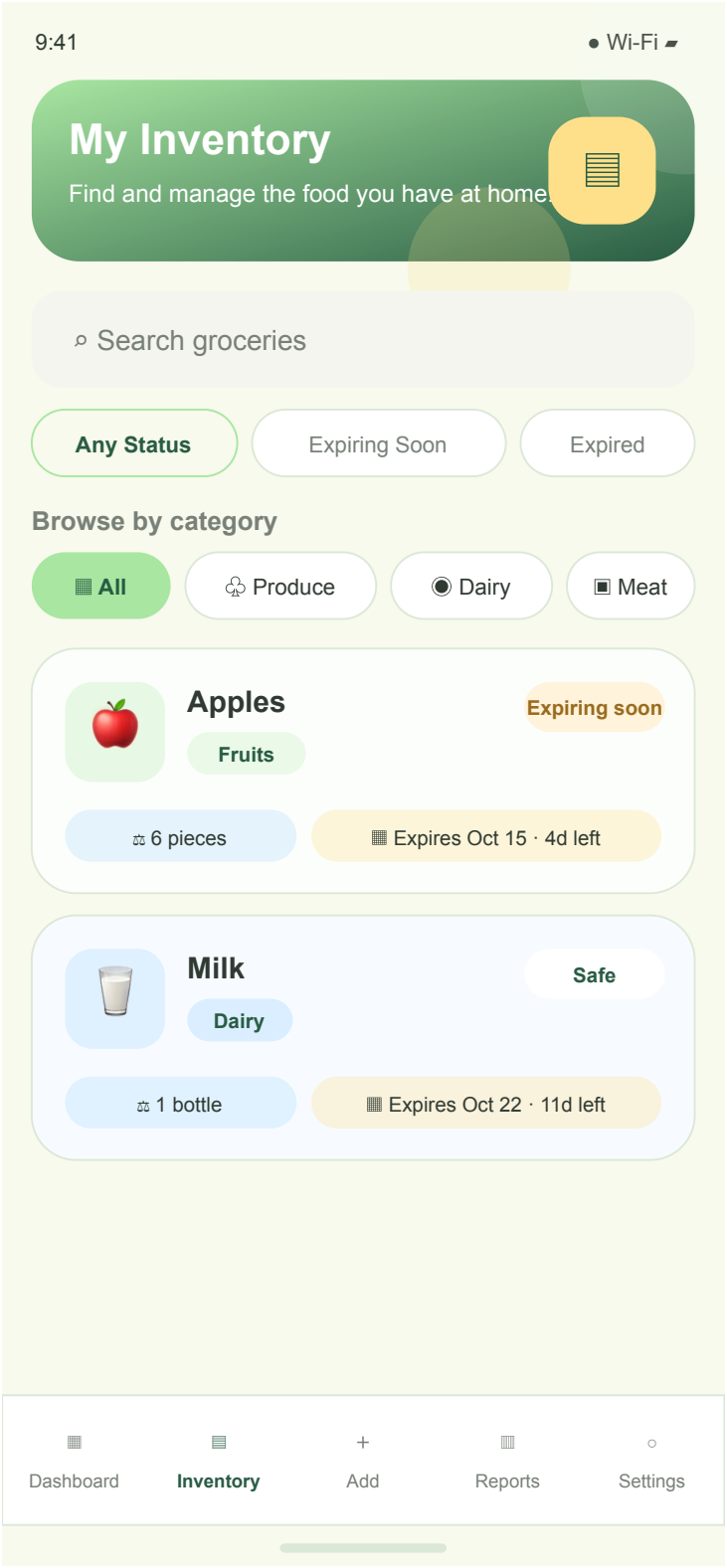


Sign in and sign up wireframe

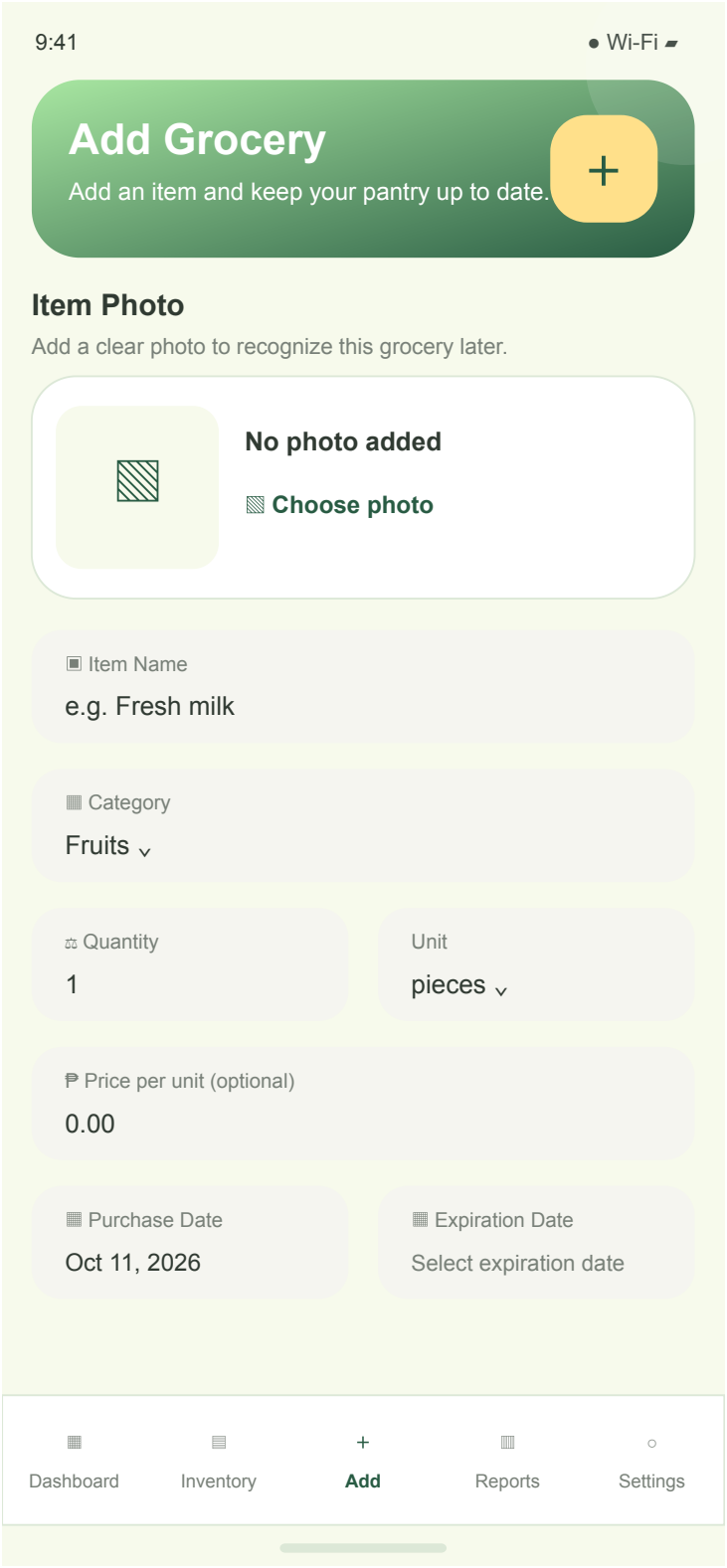
Main app



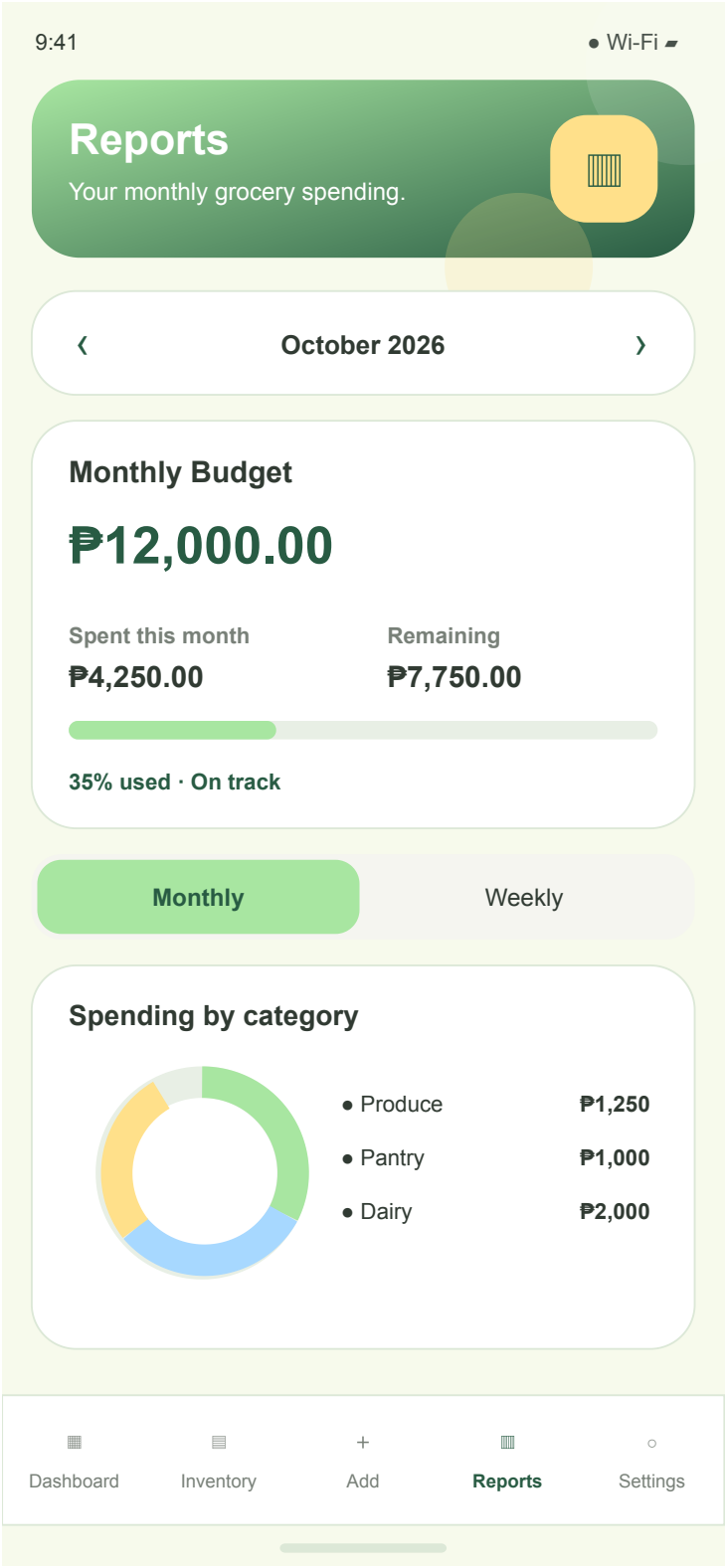
Dashboard wireframe



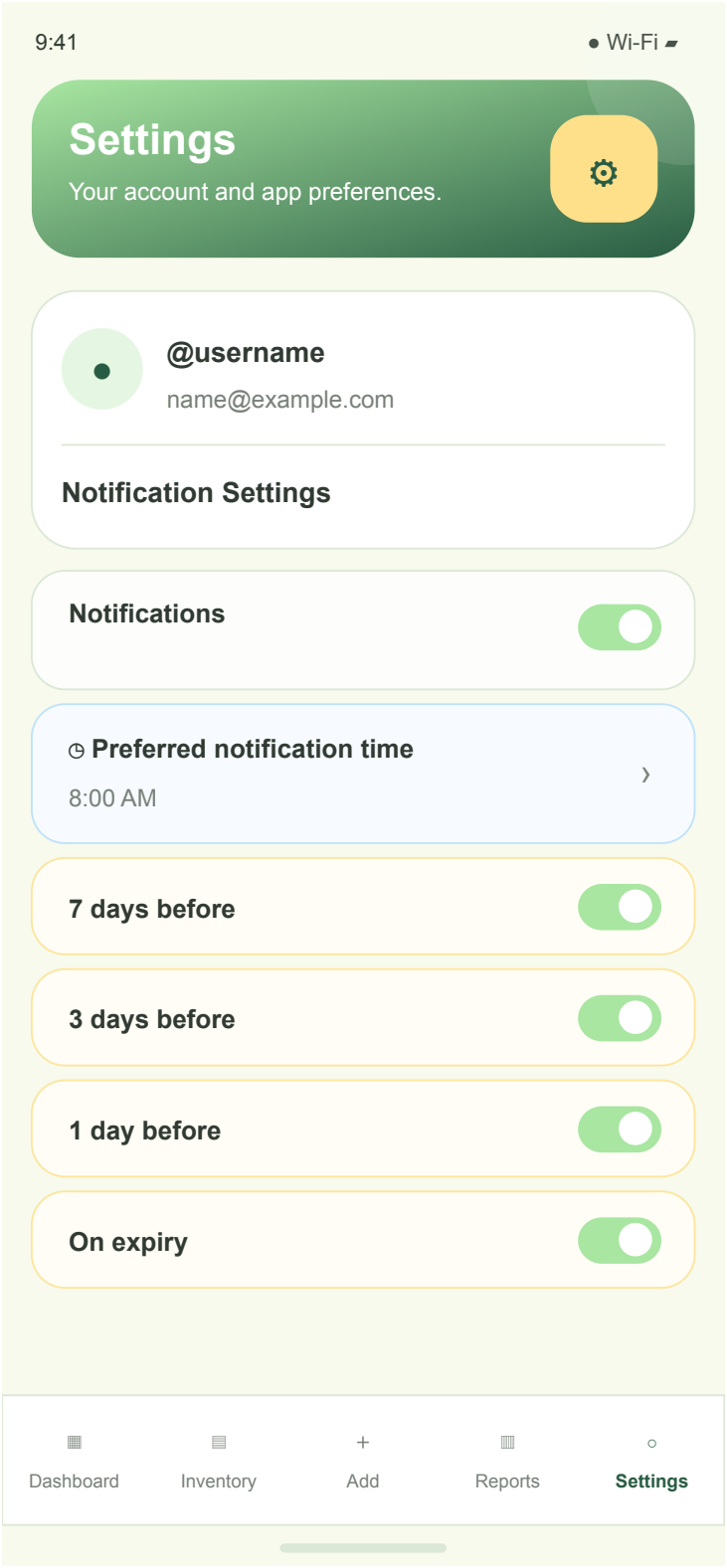
Inventory wireframe



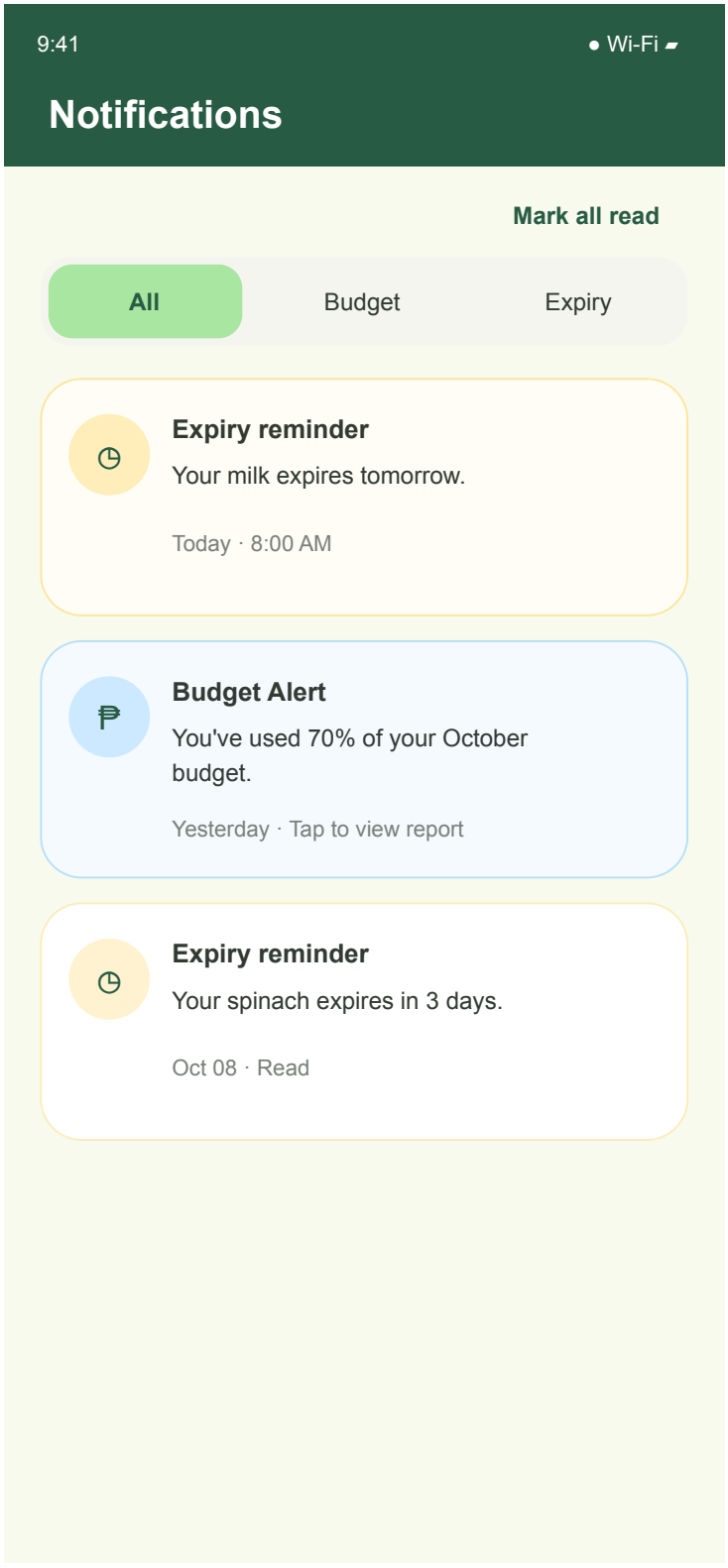
Add grocery wireframe



Reports wireframe



Settings wireframe



Notification center wireframe